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IIT-JEE ACADEMY - INDIA

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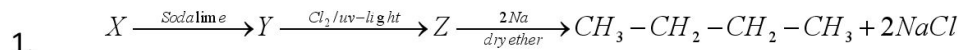


CHEMISTRY

ORGANIC CHEMISTRY PART-I(JR)
JEE ADVANCED IMPORTANT QUESTIONS

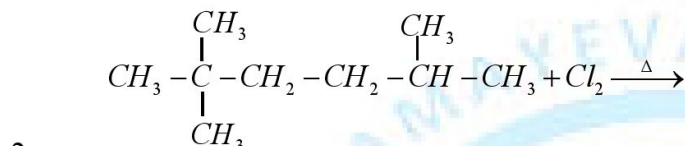
ORGANIC CHEMISTRY PART-I(JR)

SINGLE CORRECT QUESTIONS



Then X, Y and Z includes

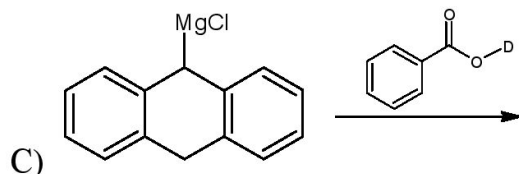
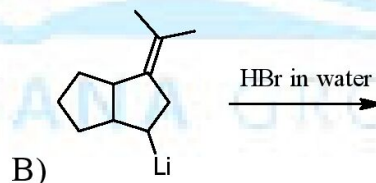
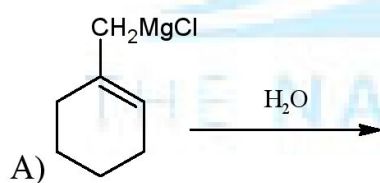
- A) Sodium ethanoate, ethane and ethyl chloride
 B) Sodiumpropanoate, ethane and methyl chloride
 C) Sodium butanoate, ethane and ethyl chloride
 D) Sodium propanoate, ethane and ethyl chloride



Major product

- A)
$$\text{CH}_3-\overset{\overset{\text{CH}_2-\text{Cl}}{|}}{\underset{\underset{\text{CH}_3}{|}}{\text{C}}}-\text{CH}_2-\text{CH}_2-\overset{\overset{\text{CH}_3}{|}}{\text{CH}}-\text{CH}_3$$
- B)
$$\text{CH}_3-\overset{\overset{\text{CH}_3}{|}}{\underset{\underset{\text{CH}_3}{|}}{\text{C}}}-\text{CH}_2-\underset{\underset{\text{Cl}}{|}}{\text{CH}}-\overset{\overset{\text{CH}_3}{|}}{\text{CH}}-\text{CH}_3$$
- C)
$$\text{CH}_3-\overset{\overset{\text{CH}_3}{|}}{\underset{\underset{\text{CH}_3}{|}}{\text{C}}}-\text{CH}_2-\text{CH}_2-\underset{\underset{\text{Cl}}{|}}{\text{C}}-\text{CH}_3$$
- D)
$$\text{CH}_3-\overset{\overset{\text{CH}_3}{|}}{\underset{\underset{\text{CH}_3}{|}}{\text{C}}}-\text{CH}_2-\text{CH}_2-\underset{\underset{\text{Cl}}{|}}{\text{C}}-\text{CH}_3$$

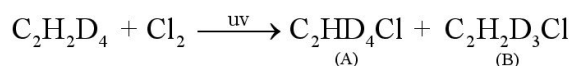
3. Which of the following halogenations reaction of alkanes is reversible?
 A) Bromination B) Chlorination C) Iodination D) Fluorination
4. In which of the following acid – base reaction takes place?



D) All of these

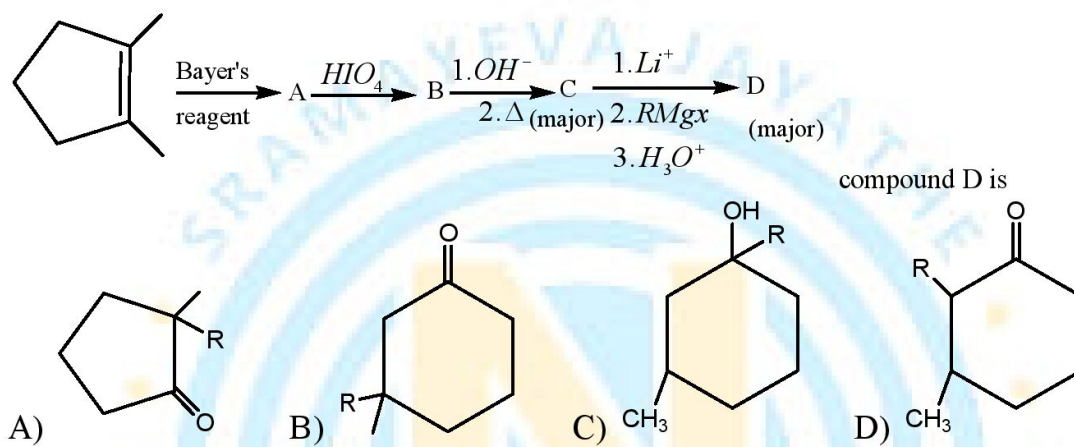
5. Reactivity of hydrogen atoms attached to different carbon atoms in alkanes has the order (towards electrophilic substitution reaction)
 A) Tertiary > Primary > secondary B) Primary > Secondary > Tertiary
 C) Both (a) and (b) D) Tertiary > secondary > Primary

6. How many isomers [including stereo isomers] (of formula $C_5H_{10}Cl_2$) are obtained from the monochlorination of (S) – (+) – 1 – chloro – 2 – methyl butane
 A) six B) five C) two D) seven
7. The relative rate of extraction of H and D is 2.6 : 1. The percentage yield of the product A and B in the following reaction is:

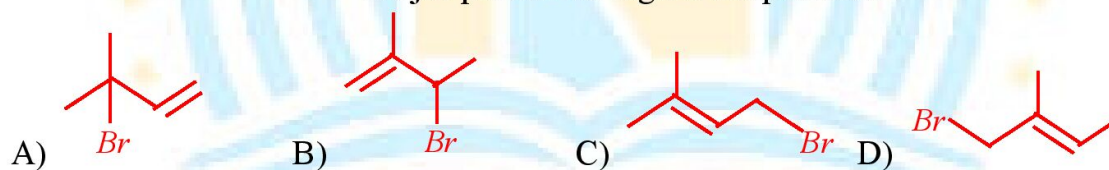


- A) 27.8%, 72.2% B) 72.2%, 27.8%
 C) 56.52%, 43.48% D) 43.48%, 56.5

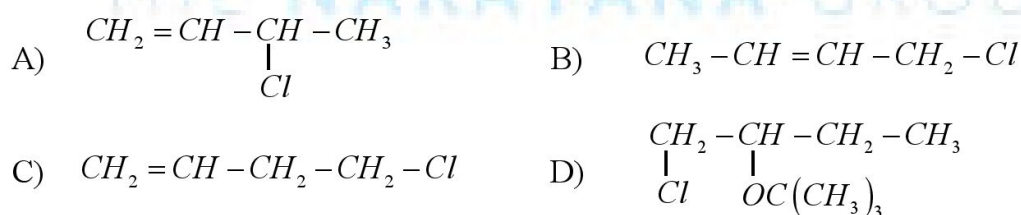
8.



9. Major product at higher temperature



10. $CH_2=CH-CH_2-CH_3 + (CH_3)_3CCOCl \xrightarrow{hv} P(Major)$

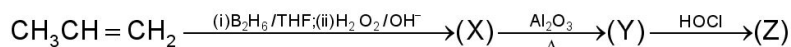


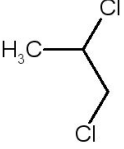
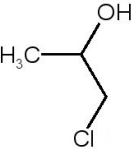
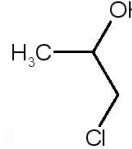
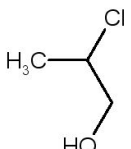
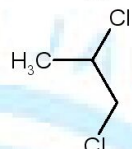
11. For the reaction of HCl with the following alkenes, predict the correct sequence of reactivity as measured by reaction rates :



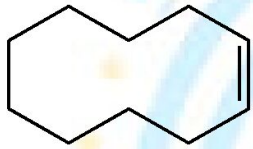
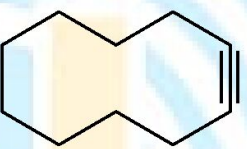
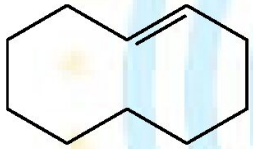
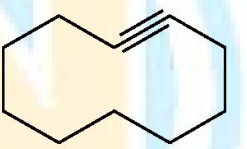
- (A) IV > I > III > II
(B) I > IV > II > III
(C) III > II > IV > I
(D) II > I > III > IV

12. Identify 'Z' in the following reaction series,



- A) Mixture of  and 
B) 
C) 
D) 

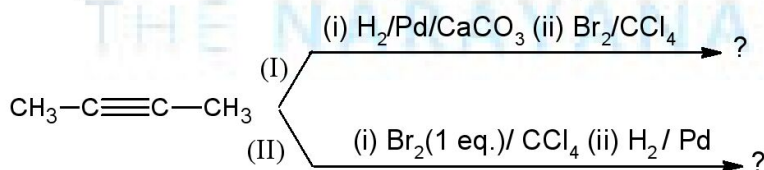
13. $\text{HC} \equiv \text{CH} \xrightarrow[\text{Cl}-(\text{CH}_2)_8-\text{Cl}]{\text{KNH}_2(2\text{ moles})} \text{A} \xrightarrow[\text{inquinoline}]{\text{H}_2, \text{Pd}-\text{BaSO}_4} \text{B} \xrightarrow{\text{Br}_2/\text{CCl}_4} \text{C} \xrightarrow[\text{(2 mole)}]{\text{KNH}_2} \text{D} \xrightarrow{\text{Na/Liq NH}_3} \text{E}$: Where E is

- A) 
B) 
C) 
D) 

14. The number of alkynes with molecular formula C_5H_8 is possible to produce metallic derivatives

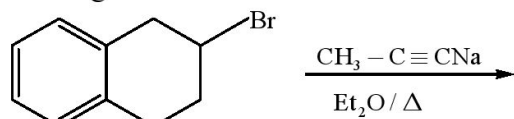
- A) 3
B) 2
C) 1
D) 0

15. The product of the following I and II sequence are related as

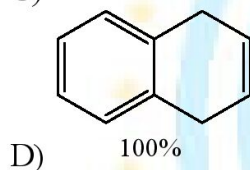
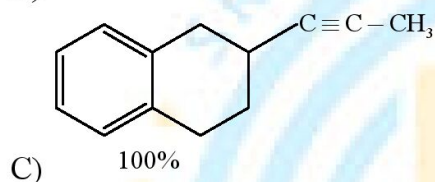
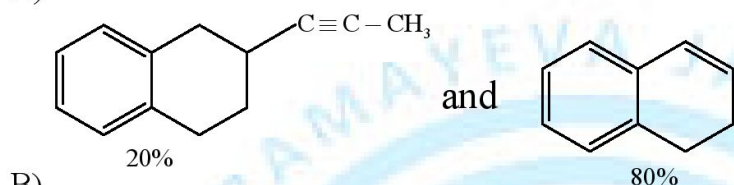
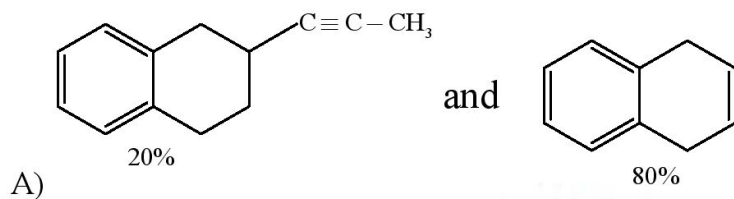


- A) Diastereomers
B) Identical
C) Enantiomers
D) Geometrical isomers

16. In the given reaction

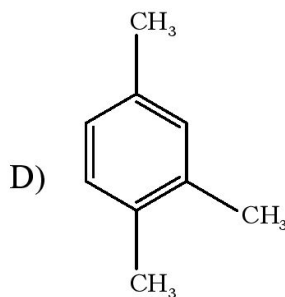
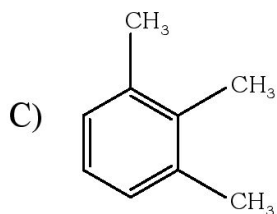
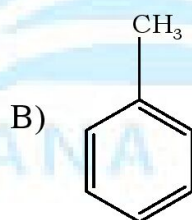
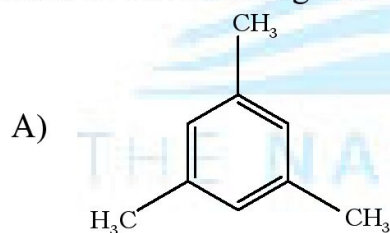


The products are

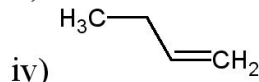
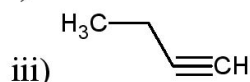
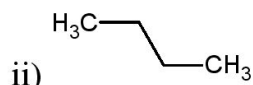
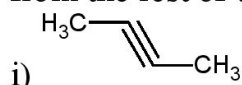


17. $\text{Me} - \text{C} \equiv \text{CH} \xrightarrow{\text{Red hot tube}} (\text{P})$

Which of the following is major product of the above reaction?



18. Which is the most suitable reagent among the following to distinguish compound (III) from the rest of the compounds



A) Br_2 in CCl_4

B) cold aq Alk. KMnO_4

C) Ammonical AgNO_3 solution

D) All of these

19. Propyne and propene can be distinguished by

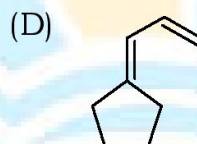
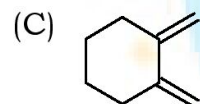
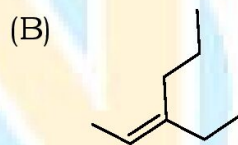
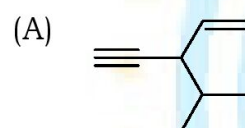
A) Conc. H_2SO_4

B) Br_2 in CCl_4

C) dil. KMnO_4

D) AgNO_3 in ammonia

20. An optically inactive hydrocarbon [A] of formula C_8H_{12} on reaction with 1 equivalent of H_2/Pd gives racemic [B] of formula C_8H_{14} . [A] reacts with ethylene at 150°C to give and optically inactive hydrocarbon [C] of formula $\text{C}_{10}\text{H}_{16}$. Reaction of [A] with NBS/light gives [D] with the formula $\text{C}_8\text{H}_{11}\text{Br}$ as a racemic mixture along with an achiral compound. Compound [A] is



21. $\text{CH} \equiv \text{CH} \xrightarrow{\text{Chromic Acid}} \text{A} \xrightarrow{\text{NaOH}} \text{B} \xrightarrow[\text{Electrolysis}]{\text{Koble's}} \text{C} \xrightarrow[\text{absence of air}]{450^\circ\text{C}} \text{D}$. Regarding 'D' the wrong statement is

A) Hybridization of carbons is sp^2

B) It is useful for the preparation of mustard gas

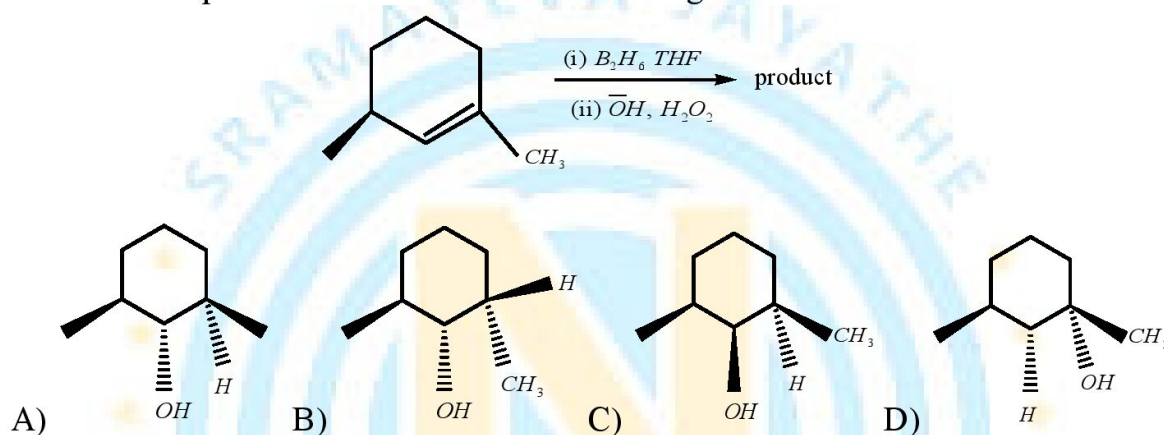
C) It gives Baeyer's test

D) It mainly participates in substitution reactions

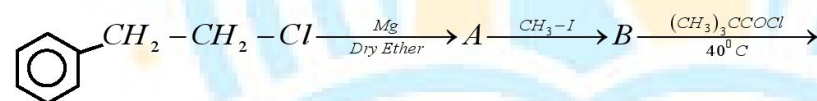
22. Two C_5H_8 isomers undergo catalytic (Pt) hydrogenation to form the same C_5H_{10} product. On ozonolysis followed by oxidative work up (H_2O_2), one isomer gave a $C_5H_8O_4$ diacid, while the other isomer gave a $C_5H_8O_3$ ketoacid. Which of the following isomeric pairs correspond to this evidence ?

- A) Cyclopentene and 1 – pentene
 B) cyclopentene and 1 – methyl cyclobutene
 C) 1 – methylcyclobutene and 3 – methylcyclobutene
 D) cyclopentene and 3 - methylcyclobutene

23. Which of the product is obtained in the following reaction :



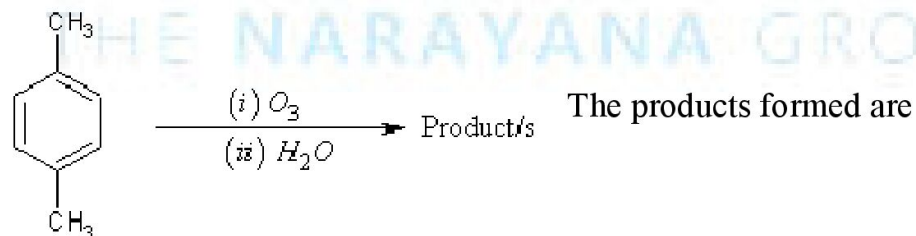
- 24.



Possible isomeric monochloro products

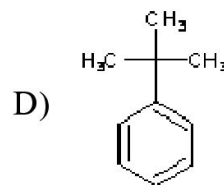
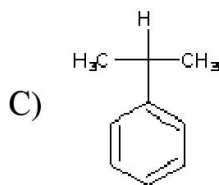
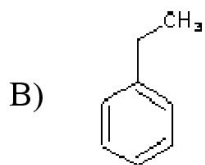
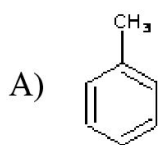
- A) 3 B) 7 C) 5 D) 6

- 25.

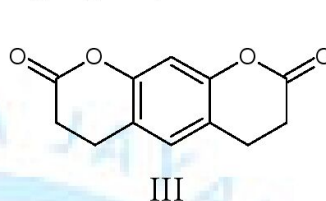
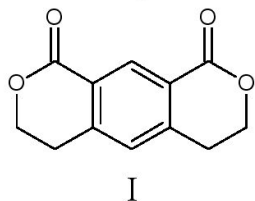


- A) $CH_3-C(=O)-COOH$ B) $HOOC-COOH$ C) $CH_3-C(=O)-C(=O)-CH_3$ D) Both (A) and (B)

26. In which of the following case, the rate of Electrophilic Aromatic Substitution will be faster.

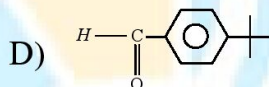
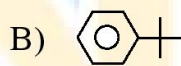
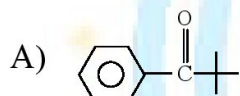
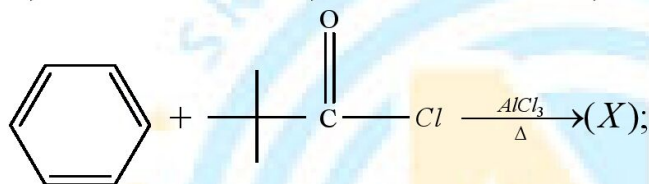


27. Increasing order of rate of reaction with $\text{HNO}_3/\text{H}_2\text{SO}_4$ is:



- A) III < II < I B) II < III < I C) I < III < II D) I < II < III

28.



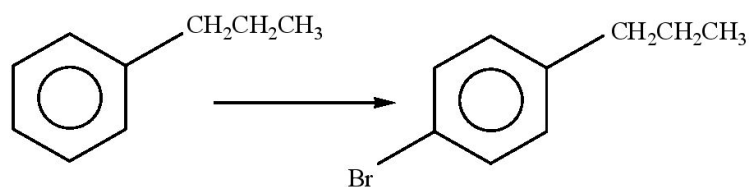
29. Write the following in decreasing order towards the electrophilic substitution reaction
(i) benzene (ii) chlorobenzene (iii) nitrobenzene (iv) toluene

- A) (i) > (ii) > (iii) > (iv) B) (iv) > (i) > (ii) > (iii)
C) (ii) > (iii) > (iv) > (i) D) (ii) > (i) > (iv) > (iii)

30. The electrophile which is considered to be the active agent in the nitration of benzene is

- A) NO_2^- B) NO^+ C) NO_2^+ D) HNO_2^+

31. The conversion



Can be best effected using

- A) Br_2 / CCl_4 B) Br_2 / H_2O C) Br_2 / Fe D) $Br_2 / Benzoyl\ peroxide$

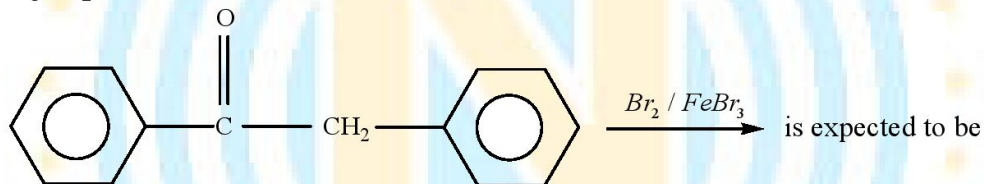
32. 1 mole of 50% pure CaC_2 on hydrolysis followed by passing through red hot copper tube, how many moles of Benzene is formed

- A) $\frac{1}{2}$ B) $\frac{1}{5}$ C) $\frac{1}{6}$ D) $\frac{1}{8}$

33. A Friedel-Crafts reaction of excess benzene with chloroform produces

- A) $C_6H_5CHCl_2$ B) $C_6H_5-\overset{\overset{Cl}{|}}{\underset{\underset{H}{|}}{C}}-C_6H_5$ C) $C_6H_5-\overset{\overset{C_6H_5}{|}}{\underset{\underset{H}{|}}{C}}-C_6H_5$ D) all of these

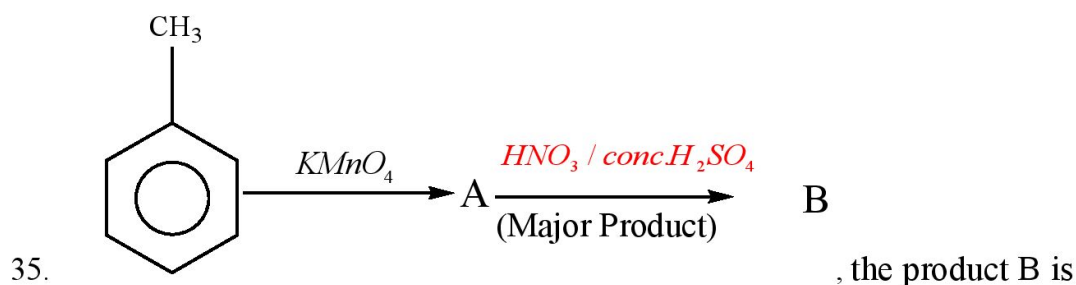
34. The major product obtained in the reaction



- A)
- B)
- C)
- D)

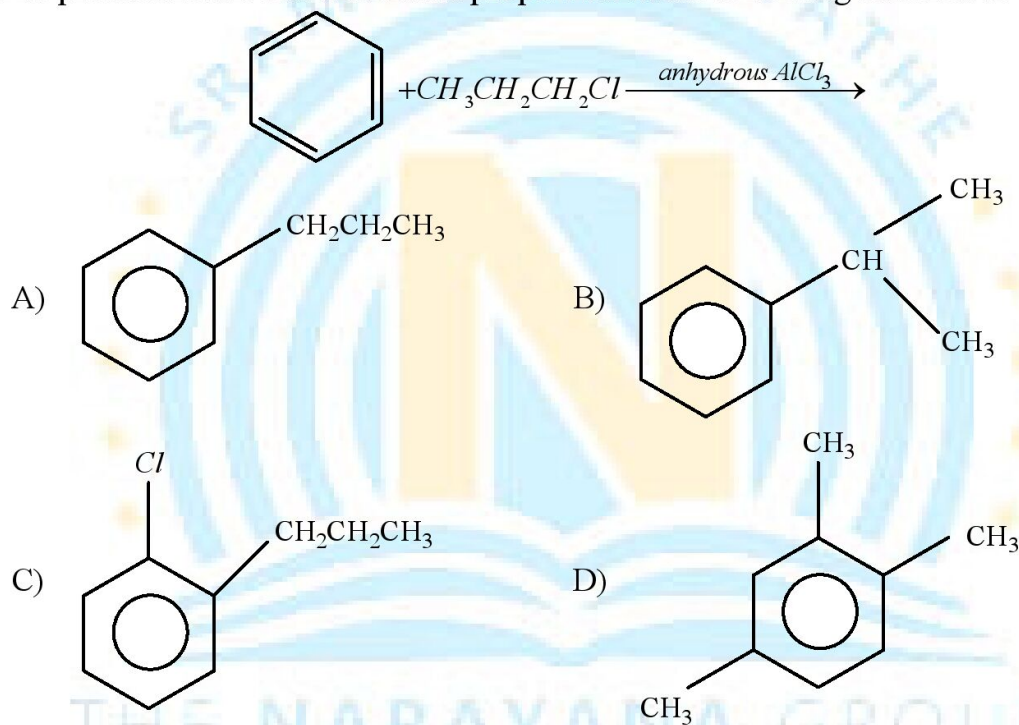
(A) (B) (C) (D)

(A) (B) (C) (D)

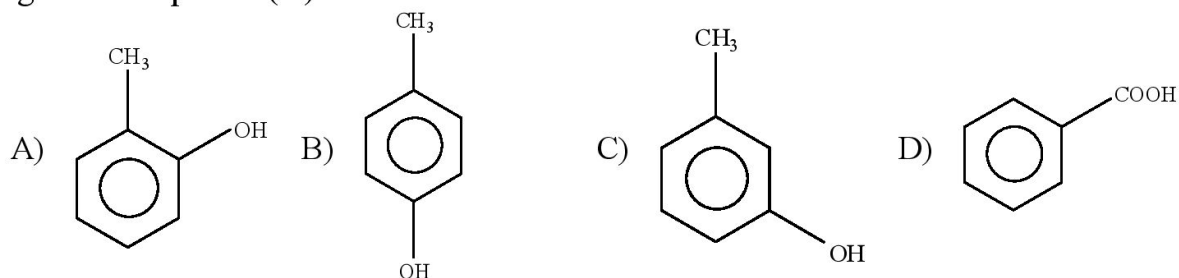


- A) 3-Nitrobenzoic acid B) 3-Nitrotoluene
C) 4-Nitrotoluene D) 4-Nitrobenzoic acid

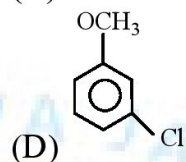
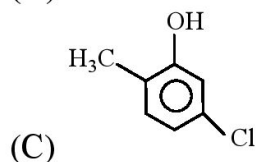
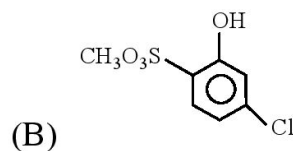
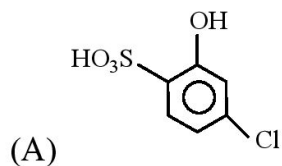
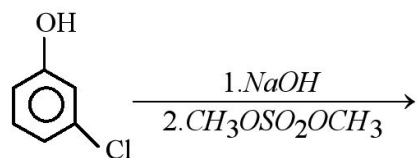
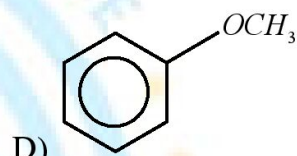
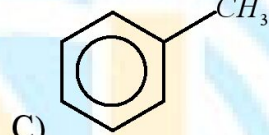
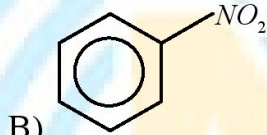
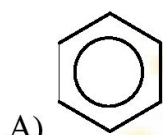
36. The product formed in maximum proportion in the following reaction is



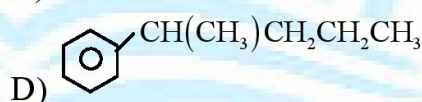
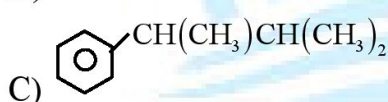
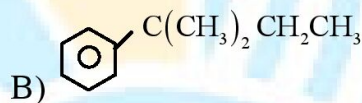
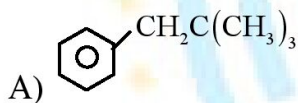
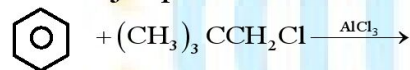
37. Chlorination of Toluene with an excess of Cl_2 in the presence of light and heat gives product (X). The compound (X) is heated with Aqueous NaOH and finally acidified to give a compound (Y). Y is



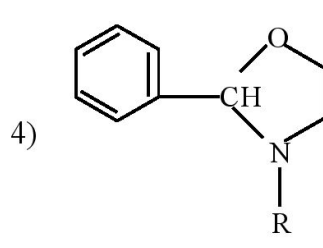
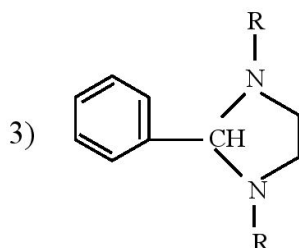
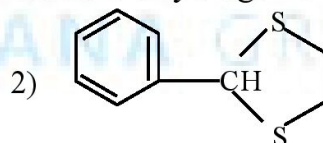
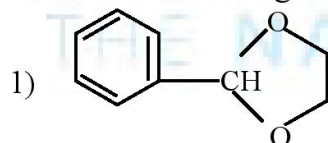
38.

39. Which of the following will have fastest rate of reaction with $\text{Br}_2 / \text{FeBr}_3$?

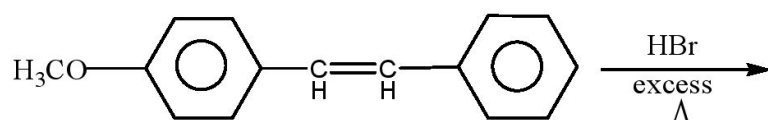
40. The major product formed in this reaction is



41. Which of the following compounds has most acidic hydrogen?

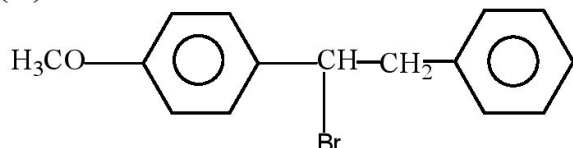


42.

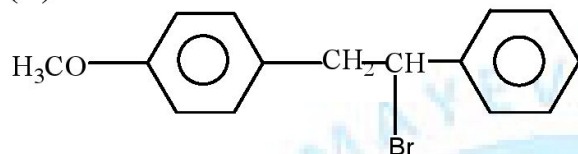


What is the product of above reaction from following option

(A)



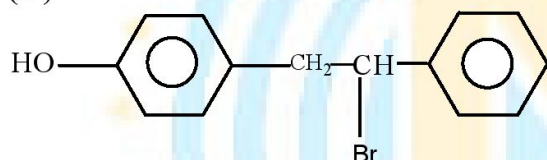
(B)



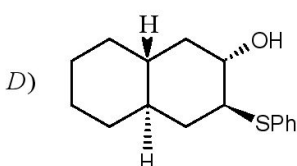
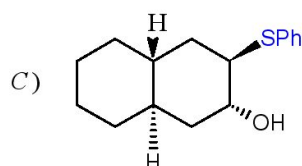
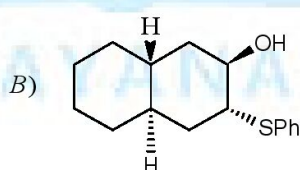
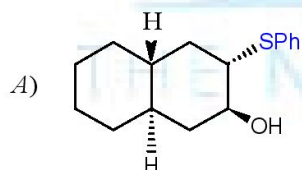
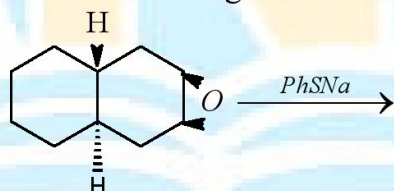
(C)



(D)



43. The major product formed in the following reaction is:



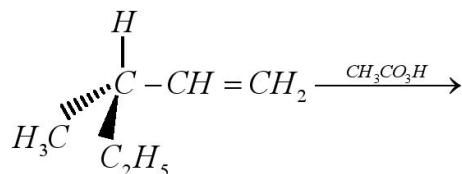
44. Consider the epoxidation of cis-butene with peroxy benzoic acid which statement describes the stereochemical out come of the reaction?

A) Cis-1,2 dimethyloxirane, mostly as (s,s) enantiomers

B) Cis-1,2 dimethyloxirane, mostly (R,S) enantiomers

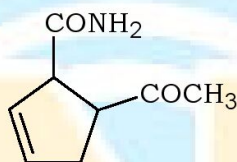
- C) Cis-1,2-dimethyloxirane, a racemate
 D) Cis-1,2-dimethyl oxirane, a meso compound

45. The reaction sequence



will yield

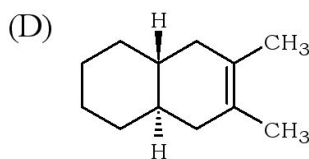
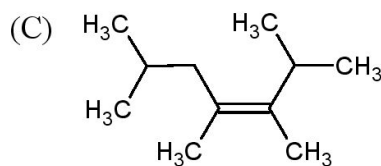
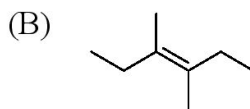
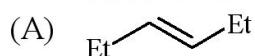
- A) A pair of products that are enantiomers
 B) A single product that is optically active
 C) A pair of products that are diastereomers
 D) A Meso compound
46. IUPAC name of the following compound is



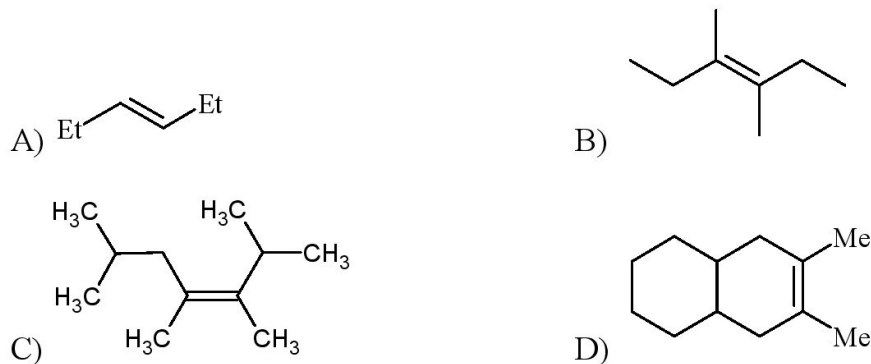
- (A) 5-acetylcyclopent-2-ene-1-carboxamide
 (B) 2-acetylcyclopent-4-ene-1-carboxamide
 (C) 1-(2-carbamoylcyclopent-3-enyl)ethan-1-one
 (D) 5-acetylcyclopent-2-en-1-carboxamide
47. The C – H bond having the lowest dissociation energy is
- (A) $\text{PhH}_2\text{C} - \text{H}$ (B) $\text{H}_2\text{C}=\text{CH} - \text{CH}_2 - \text{H}$
 (C) $\text{Me}_2\text{HC} - \text{H}$ (D) $\text{Me}_3\text{C} - \text{H}$

MULTI CORRECT QUESTIONS

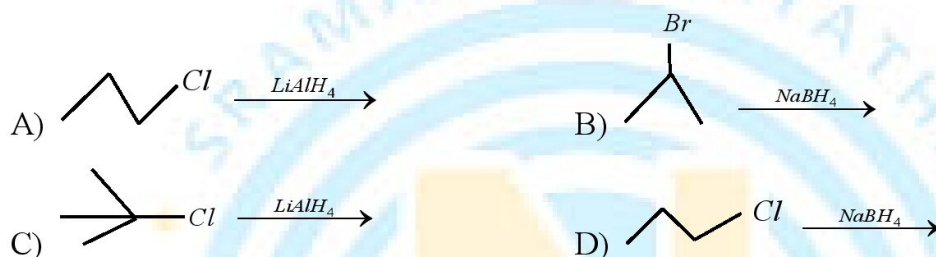
48. Which of the following hydrocarbons give *dl*-pair on catalytic hydrogenation to alkanes?



49. Which of the following hydrocarbons give dl-pair on catalytic hydrogenation to alkanes?



50. Which of the following reactions will give Alkane as major product?



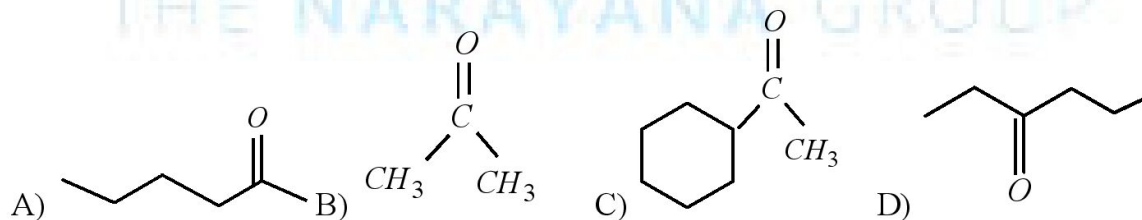
51. An alkene on reductive ozonolysis yields only ethanal. The other isomer(s) of this alkene which on reductive ozonolysis may give

(A) propanal (B) methanal (C) propanone (D) isobutanal

52. The alkynes $\text{CH}_3\text{--CH}_2\text{--C}\equiv\text{CH}$ and $\text{CH}_3\text{--C}\equiv\text{C--CH}_3$ can be distinguished by the following

(A) Tollen's Reagent (B) Ammoniacal Cu_2Cl_2 solution
(C) Fehling's solution (D) Baeyer's reagent

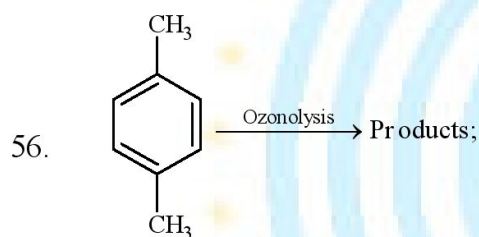
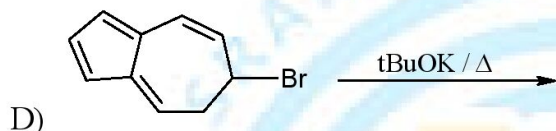
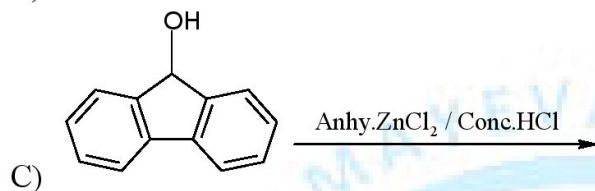
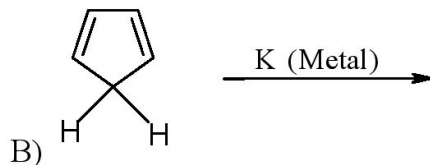
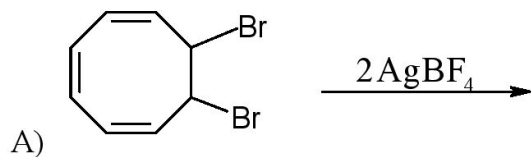
53. Which of the following are obtained as the only product after hydration of suitable alkyne with H_2SO_4 , and HgSO_4 ?



54. Which of the following reaction will be useful for conversion of P_3 to P_4 .

A) Clemmenson's reduction (B) Oppenauer oxidation
(C) Meerwin Ponderff verely reduction (D) Tischenko reaction

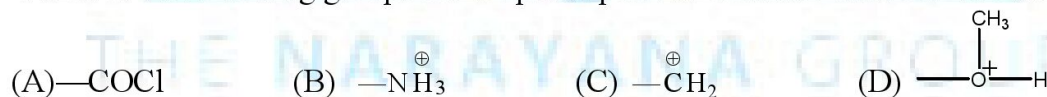
55. Which of the following reaction(s) give(s) aromatic compound as product?



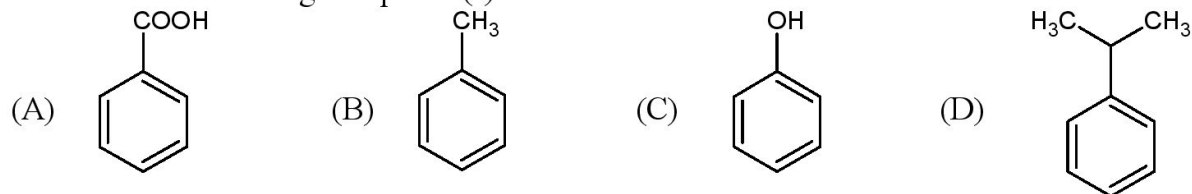
The products formed are :



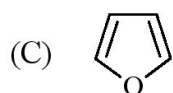
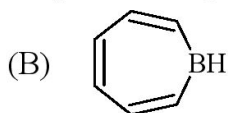
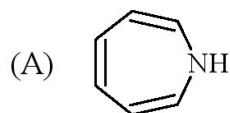
57. Which of the following groups cannot participate in resonance with benzene ring?



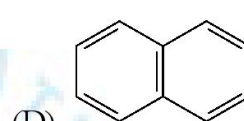
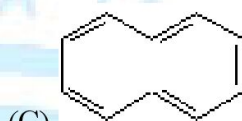
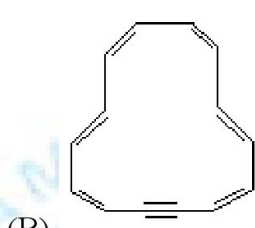
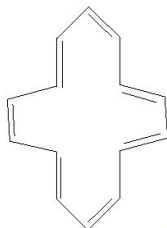
58. Which of the following compound(s) is/are more reactive than benzene towards nitration?



59. Which of the following heterocyclic compound/s might be aromatic?



60. Among the following, aromatic compounds are



61. In which of the following reaction t-butylbenzene is formed?

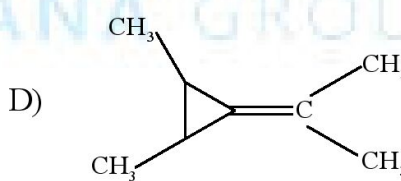
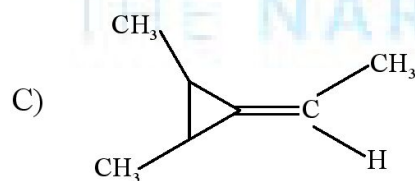
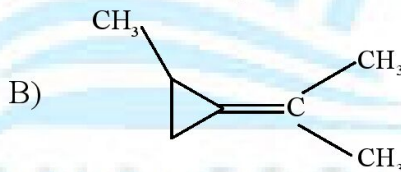
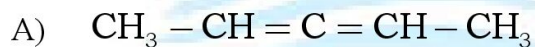
A) Benzene + iso-butyl chloride, $AlCl_3$

B) Benzene + $(CH_3)_2C=CH_2 \xrightarrow{BF_3 \cdot HF}$

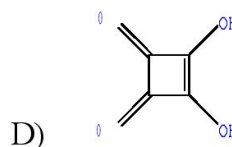
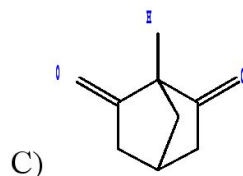
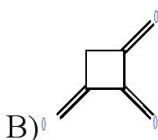
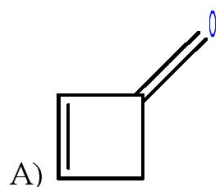
C) Benzene + t-butyl alcohol $\xrightarrow{H_2SO_4}$

D) Benzene + $(CH_3)_2C=CH_2 \xrightarrow{AlCl_3}$

62. Which of the following will show optical isomerism as well as geometrical isomerism

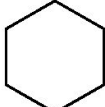
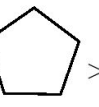


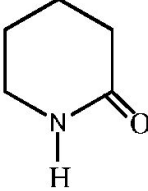
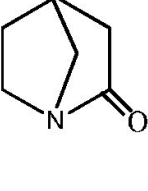
63. Which of the following compound will not show enolisation

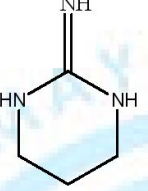
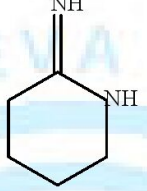
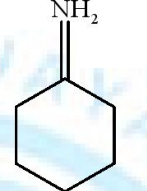


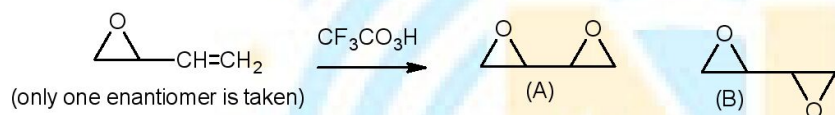
64. Select the correct order among the following:

A) Halide nucleophilicity in DMSO solvent is $F^- > Cl^- > Br^- > I^-$

B) Order of heat of combustion  $>$  $>$ 

C)  $<$ 
Order of basic strength is

D)  $>$  $>$ 
Order of basic strength



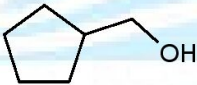
Which of the following statements is incorrect about A and B ?

(A) A and B are mixture of diastereomers

(B) A and B are mixture of enantiomers

(C) A and B are optically active

(D) B is optically active and A is optically inactive

66. The alcohol  on dehydration gave three isomeric alkenes **A**, **B** and **C**. one mole of each of these are subjected to ozonolysis. Then **A** gave one mole of dialdehyde **B** gave one mole of an aldehyde and one mole of ketone whereas **C** gave one mole of keto aldehyde. The correct statement(s) from the following is/are

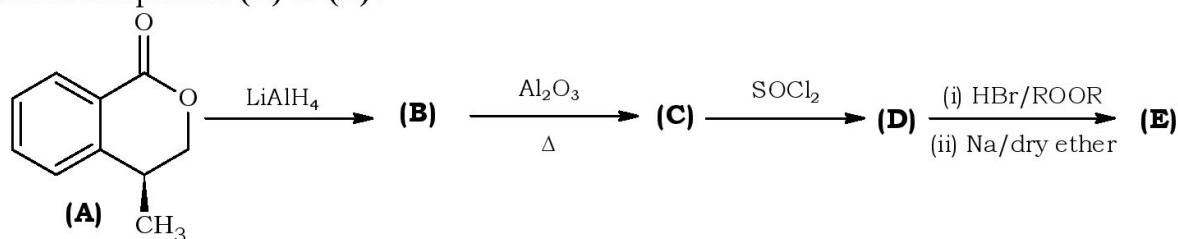
(A) In the formation of **A** from the reactant ring expansion takes place.

(B) In the formation of **B** from the reactant 1, 2-hydride shift takes place

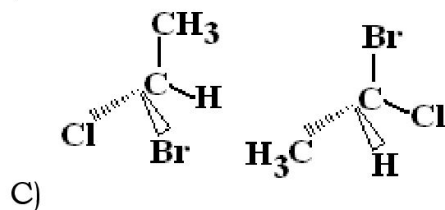
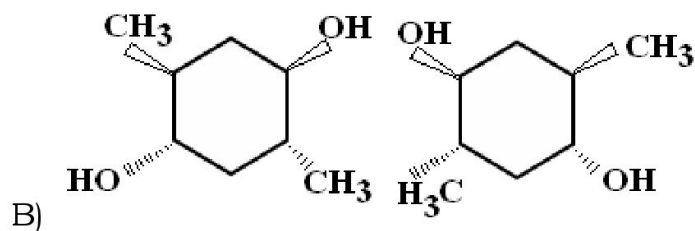
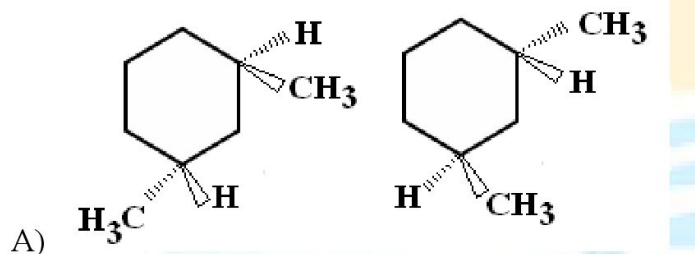
(C) **B** is Hoffmann product

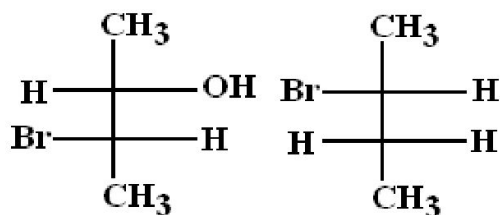
(D) **C** is Zaitsev product.

67. With reference to the scheme given below, which of the following statement(s) is/are true about compounds (A) to (E)?



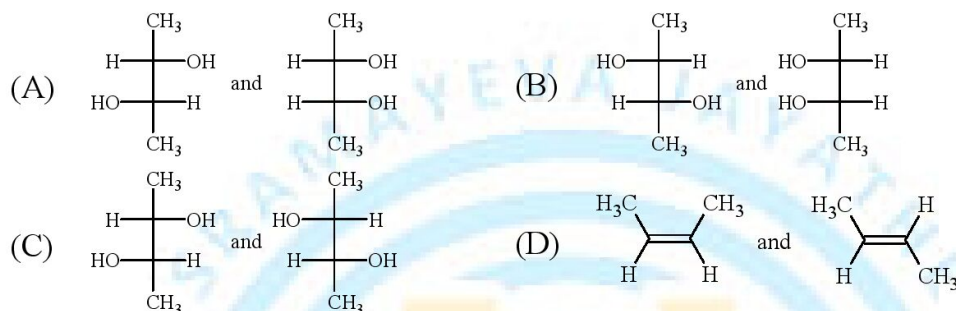
- (A) Compound (B) releases a gas on treatment with sodium metal.
- (B) Compound (C) gives a resolvable mixture on reaction with $\text{Hg}(\text{OAc})_2$ followed by reduction with NaBH_4 .
- (C) Compound (D) does not exhibit geometrical isomerism.
- (D) (E) is a mixture containing achiral molecules.
68. The different kinds of isomerisms exhibited by the compound $\text{CH}_3\text{COCH}=\text{CHCH}(\text{Cl})\text{CH}_3$ is/are
- (A) Optical isomerism (B) Geometrical isomerism
- (C) Tautomerism (D) Conformational isomerism
69. Which one of the following pairs of isomers are enantiomers





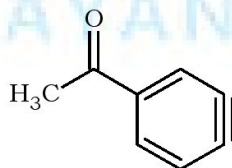
D)

70. Which of the following pairs of compounds are diastereomers?

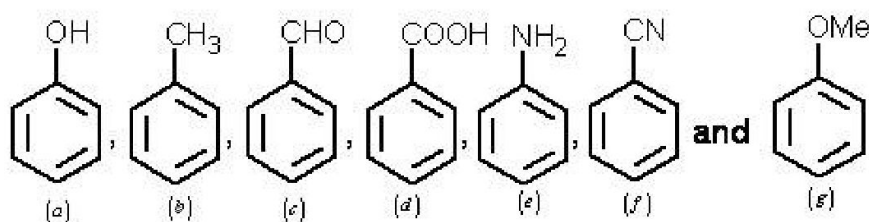


NUMERICAL TYPE QUESTIONS

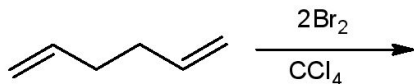
71. In the monochlorination of 3-methylpentane, let x be the number of pairs of isomers which exist as enantiomers, y be the number of pairs of isomers which exist as diastereomers, z be the number of isomers which are achiral. Calculate the value of $(x + y + z)$
72. An alkene C_6H_{12} (A) undergoes oxidative ozonolysis to give two different acids (B) and (C). Neither (B) nor (C) contains more than four carbon atoms. How many alkenes can satisfy this criteria?
73. The number of alkenes having higher heat of combustion than but-1-ene is
Ethene, propene, *cis* - but-2-ene, *trans* - but-2-ene, pent-1-ene, 2-methylprop-1-ene
74. The number of sp^2 carbon atoms in the following molecule is



75. How many of the following groups are activating towards electrophilic aromatic substitution?
 $-\text{NHCOR}$, $-\text{OCOR}$, $-\text{C}(=\text{O})-\text{O}-\text{R}$, $-\text{NR}_2$, $-\text{NH}_2$, $-\text{OH}$, $-\text{OR}$
76. Number of different products formed in mononitration of 1, 3 - dibromo benzene including minor products is
77. How many of the following compounds are more reactive than benzene for EAS reaction ?



78. Total number of stereoisomeric tetrabromides that will be formed in the following reaction is



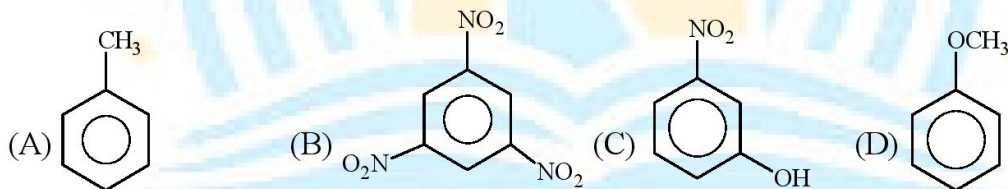
79. How many saturated hydrocarbons can be prepared from alkenes of formula C_6H_{12} excluding stereoisomers? (None of the structures contain quaternary carbon).

PARAGRAPH TYPE QUESTIONS

Q.NO: 80-81

Electrophilic substitutions are characteristic reaction of aromatic compounds. In this reaction hydrogen of the aromatic ring is replaced by an electrophile because hydrogen in the form of H^+ is very good leaving group. $-\text{COOH}$ and $-\text{SO}_3\text{H}$ are also good leaving groups when they are present either at *ortho* or at *para* to the highly activating groups like $-\text{NH}_2$ and $-\text{OH}$.

80. Which one of the following compounds will be **most** unreactive towards electrophilic substitution reaction?



81. Benzene sulphonic acid is converted into benzene by
 (A) heating with dilute sulphuric acid (B) heating with dilute NaOH
 (C) heating with soda lime (D) heating with Zinc dust

Paragraph for Questions 82 and 83

Two compounds A and B of same molecular formula C_4H_6 gave the following reactions.

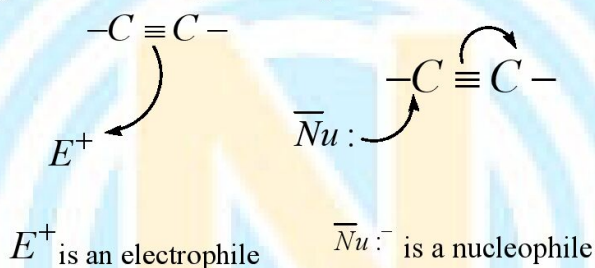
- Both discharged colour of Br_2 in CCl_4
- (A) gives a precipitate with ammonical AgNO_3 and on oxidation gives propanoic acid and CO_2 .
- (B) does not produce precipitate with ammonical AgNO_3 and on oxidation gives oxalic

acid and CO_2 .

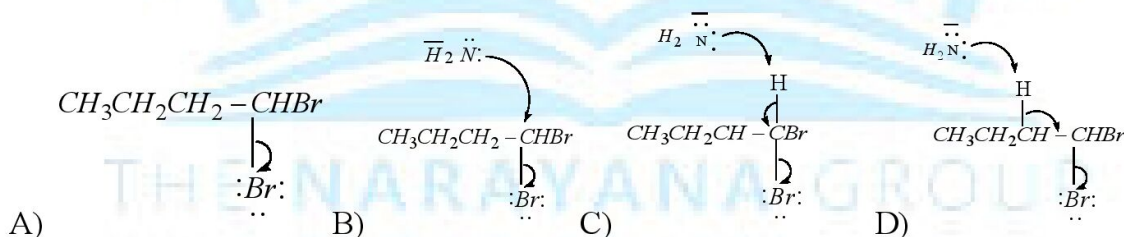
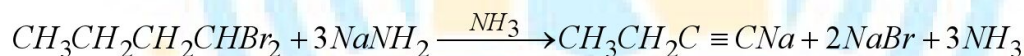
82. The compound (A) is
 (A) $\text{H}_3\text{C}-\text{CH}_2-\text{CH}\equiv\text{CH}$ (B) $\text{H}_2\text{C}=\text{CH}-\text{CH}=\text{CH}_2$
 (C) $\text{H}_3\text{C}-\text{C}\equiv\text{C}-\text{CH}_3$ (D) 
83. The compound (B) is
 (A) $\text{H}_2\text{C}=\text{C}=\text{CH}-\text{CH}_3$ (B) $\text{H}_2\text{C}=\text{CH}-\text{CH}=\text{CH}_2$
 (C) $\text{H}_3\text{C}-\text{C}=\text{C}-\text{CH}_3$ (D) 

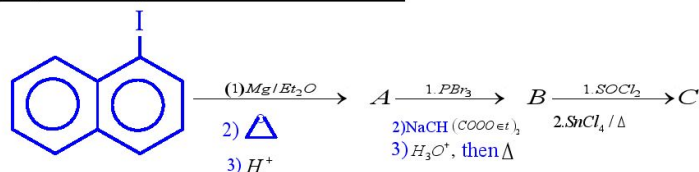
Paragraph for Questions 84 and 85

84. The $\equiv\text{C}-\text{H}$ unit of an alkyne is more acidic than a $\text{C}-\text{H}$ unit of an alkene or alkane, allowing acetylene and terminal alkynes to be converted to their conjugate bases $\equiv\text{C}^-$
 2 Unlike alkenes, alkynes are reduced by metals, especially *Li*, *Na*, and *K*
 3 Unlike alkenes, alkynes can undergo nucleophilic as well as electrophilic addition.

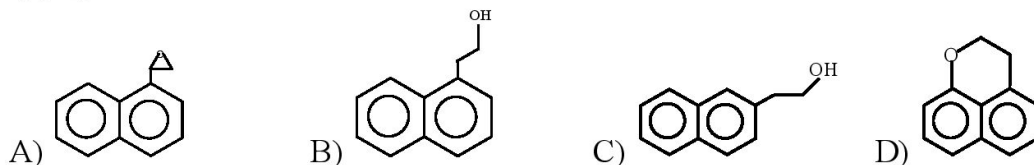


85. Which of the following best describes what happens in the first step in the mechanism of the reaction shown?



Paragraph for Questions 86 and 87

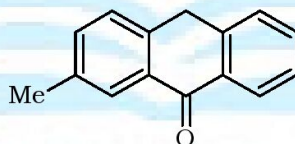
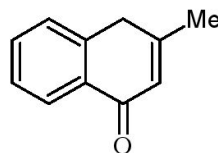
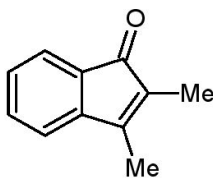
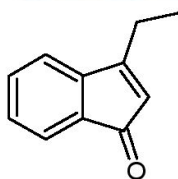
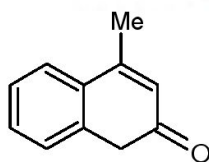
86. "A" is



87. Product "C" is

Paragraph for Questions Nos. 88 to 90

A hydrocarbon **P** ($C_{11}H_8$) on treatment with dilute H_2SO_4 in presence of $HgSO_4$ forms **Q** ($C_{11}H_{12}O_2$). **P** on treatment with $B_2H_6/H_2O_2 - NaOH$ gives **R** which is an isomer of **Q**. **Q** gives yellow precipitate with $NaOI$ while **R** doesn't. **R** forms a grey precipitate on treatment with ammoniacal silver nitrate solution. **P** on boiling with alkaline solution of $KMnO_4$ forms $C_8H_6O_4$, which on heating with P_4O_{10} yields **S** ($C_8H_4O_3$). **S** on treatment with toluene in presence of anhydrous $AlCl_3$ yields **T** ($C_{15}H_{12}O_3$) as the major product. **T** is reduced by $Zn-Hg/Con. HCl$ and product obtained gives the following compound on reaction with HF .

88. Which of the following products are formed when **Q** is treated with aqueous $NaOH$?I
IV

II

III

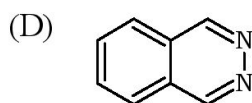
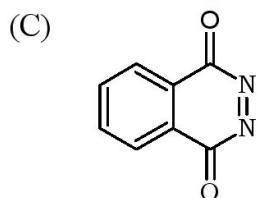
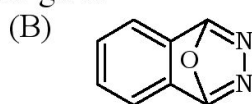
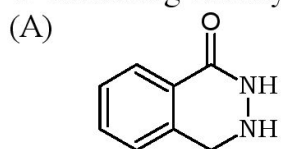
(A) I & IV

(B) I, II & III

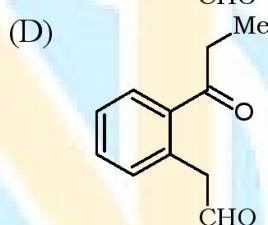
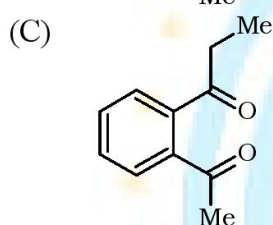
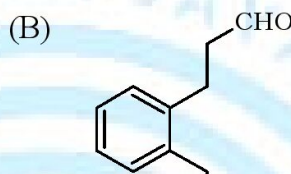
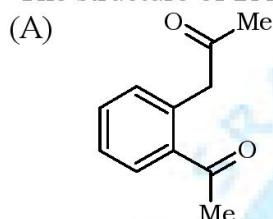
(C) II & III

(D) I, II, III & IV

89. **S** on heating with hydrazine hydrochloride gives

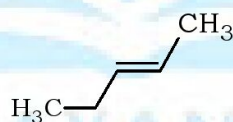


90. The structure of **R** is



TWO COLUMN MATCHING

91. Consider the compound given below. Column-I has the reagents which react with this compound. Match the reactants in Column – I with the characteristics of the reactions in Column – II.



	Column I		Column II
(A)	Dilute H_2SO_4	(P)	Produces a mixture of two different carbonyl compounds
(B)	$\text{OsO}_4/\text{H}_2\text{O}_2/\text{H}_2\text{O}$	(Q)	Produces a mixture of two pairs of enantiomers
(C)	Bromine water	(R)	Involves cyclic intermediate.
(D)	(i) O_3 (ii) $\text{Zn}/\text{CH}_3\text{COOH}$	(S)	Product contains at least one alcohol group

		(T)	Product on reaction with concentrated H_2SO_4 produces a mixture of the reactant and its diastereomer.
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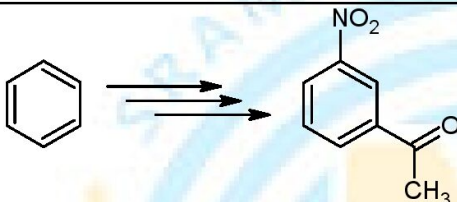
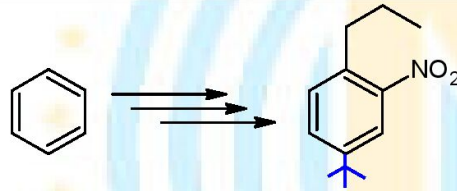
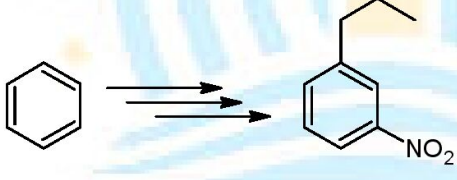
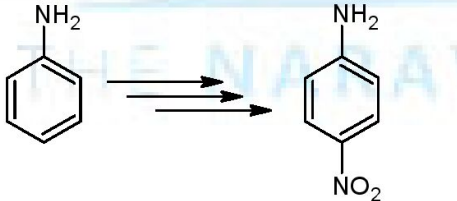
(A) A-ST;B-RS;C-RS;D-PR

(B) A-RS;B-RT;C-PR;D-QR

(C) A-PT;B-QS;C-PT;D-QT

(D) A-QT;B-TS;C-QS;D-PQ

92. Match the multistep conversions in Column – I with appropriate sequence of reagents given in Column – II.

	Column I		Column II
(A)		(P)	(i) Ac_2O (ii) $\text{NO}_2^+\text{BF}_4^-$ (iii) NaOH/Δ
(B)		(Q)	(i) $\text{Con. HNO}_3/\text{Con. H}_2\text{SO}_4$ (ii) $\text{Ac}_2\text{O}/\text{anhyd. AlCl}_3$
(C)		(R)	(i) $\text{CH}_3\text{COCl}/\text{anhyd. AlCl}_3$ (ii) $\text{Con. HNO}_3/\text{Con. H}_2\text{SO}_4$
(D)		(S)	(i) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}/\text{anhyd. AlCl}_3$ (ii) $t\text{-BuCl}/\text{anhyd. AlCl}_3$ (iii) $\text{Con. HNO}_3/\text{Con. H}_2\text{SO}_4$
		(T)	(i) $\text{CH}_3\text{CH}_2\text{COCl}/\text{anhd. AlCl}_3$ (ii) $\text{Con. HNO}_3/\text{Con. H}_2\text{SO}_4$ (iii) $\text{NH}_2\text{NH}_2/\text{NaOH}/\text{glycol}$

- (A) A – S; B – R; C – T; D – Q (B) A – R; B – S; C – T; D – P
 (C) A – Q; B – R; C – S; D – T (D) A – Q; B – P; C – S; D – T

93. Match List-I with List-II and select the correct options given below:

List-I (Compound)**List-II (Property)**

- (P) Dihydro furan (1) Non-planar structure
 (Q) Borazine (2) Antiaromatic
 (R) Cyclooctatetraene (3) Aromatic
 (S) [12] Annulene (4) Five membered ring

Codes:

- | | P | Q | R | S |
|-----|---|---|---|---|
| (A) | 2 | 3 | 1 | 4 |
| (B) | 4 | 1 | 2 | 3 |
| (C) | 4 | 3 | 1 | 2 |
| (D) | 4 | 3 | 2 | 1 |

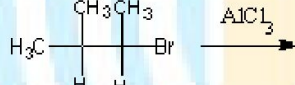
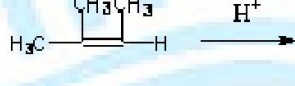
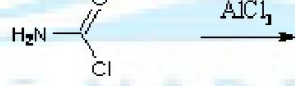
94. Match the column:

	List – I		List – II
(P)	Propanenitrile+MeMgBr $\xrightarrow[2.H_3O^+]{1.THF,\Delta}$	(1)	
(Q)	Propanoylchloride+2MeMgBr $\xrightarrow[2.H_3O^+]{1.THF,\Delta}$	(2)	
(R)	Ethylethanoate+2MeMgBr $\xrightarrow[2.H_3O^+]{1.THF,\Delta}$	(3)	
(S)	Propylmethanoate+2MeMgBr $\xrightarrow[2.H_3O^+]{1.THF,\Delta}$	(4)	

Code :

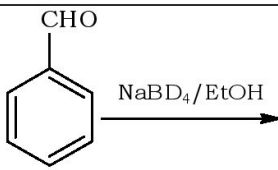
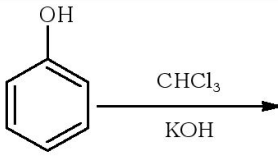
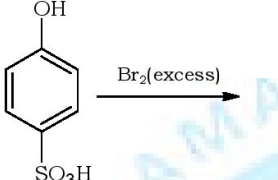
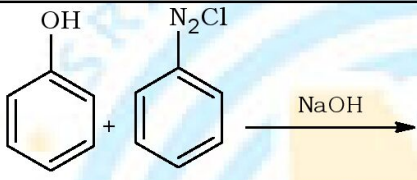
	P	Q	R	S		P	Q	R	S
(A)	1	3	4	2	(B)	2	4	3	1
(C)	3	1	2	4	(D)	3	2	1	4

95. Match the following :

	Column I(Ion)		Column II
a)	 +  $\xrightarrow{AlCl_3}$	(p)	Cation intermediate
b)	 +  $\xrightarrow{H^+}$	(q)	Rearranged product
c)	 +  $\xrightarrow{AlCl_3}$	(r)	Exhibits Isotopic(K_H/K_D) effect
d)	 $\xrightarrow{H_2S_2O_7}$	(s)	Rate of reaction increases in presence of EDG

(A) A $\rightarrow$ Q,R,S, B $\rightarrow$ R,S, C $\rightarrow$ Q,S, D $\rightarrow$ Q,R,S(B) A $\rightarrow$ R,S, B $\rightarrow$ Q,R, C $\rightarrow$ Q,S, D $\rightarrow$ P,Q,S(C) A $\rightarrow$ P,Q,S, B $\rightarrow$ P,S, C $\rightarrow$ P,S, D $\rightarrow$ P,R,S(D) A $\rightarrow$ P, S, B $\rightarrow$ Q,S, C $\rightarrow$ R,S, D $\rightarrow$ P,Q,R

96. Match the reactions in column I with appropriate properties of it in column II.

	Column I		Column II
(A)		(P)	Product exhibits stereoisomerism
(B)		(Q)	Does not involve carbocation intermediate
(C)		(R)	Electrophilic aromatic substitution
(D)		(S)	Nucleophilic addition
		(T)	Product is a <i>d,l</i> pair

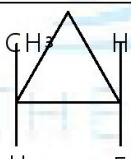
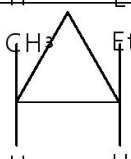
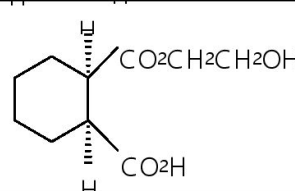
(A) A-PQT; B-RS; C-S; D-PQT

(B) A-PST; B-RS; C-T; D-QST

(C) A-QST; B-QS; C-R; D-PQS

(D) A-PQST; B-QR; C-R; D-PQR

97. Match the following :

	COLUMN - I		COLUMN - II
a)		p)	Geometrical isomerism
b)		q)	Optical isomerism
c)		r)	Tautomer

d)		s)	Meso
----	--	----	------

(A) A – S; B – R; C – Q; D – P

(B) A – R; B – R; C – P; D – S

(C) A – Q; B – P; C – R; D – S

(D) A – S; B – Q; C – P; D – R

98. Match the Following:-

	Column – I		Column – II
(A)		(p)	Optically active
(B)		(q)	Optically inactive
(C)		(r)	Plane of symmetry
(D)		(s)	Number of chiral centre = 1
		(t)	Number of of stereocentres = 2

(A) A-Q; B-QS; C-RS; D-RT

(B) A-R; B-PS; C-QR; D-QR

(C) A-S; B-RS; C-PR; D-QT

(D) A-Q; B-ST; C-RT; D-RS

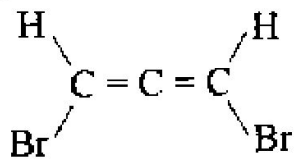
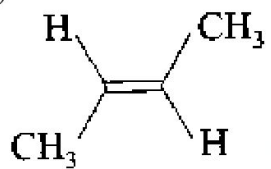
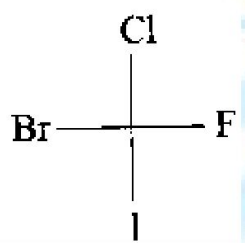
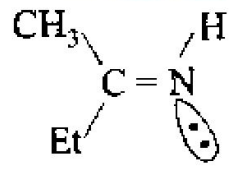
99. Match the following.

Column – I		Column – II	
(A)		(p)	18.4 – 25.5 KJ/mol
(B)		(q)	14.2 KJ/mol
(C)		(r)	3.3 KJ/mol
(D)		(s)	Zero

(A) A-PQ; B-PQR; C-PS; D-QR
 (C) A-PR; B-QRS; C-QS; D-RS

(B) A-QR; B-PQS; C-QS; D-QS
 (D) A-PS; B-QRS; C-RS; D-PR

100. Match the Column I and II.

Column – I	Column – II
a) 	p) Chiral centres containing compound
b) 	q) Presence of stereocentre
c) 	r) Optically active compound
d) 	s) Compound containing plane of symmetry

(A) A-QS; B-PQ; C-QS; D-RS

(B) A-PR; B-QR; C-RS; D-PQ

(C) A-QR; B-PR; C-QR; D-PR

(D) A-PQ; B-QR; C-RS; D-PR

KEY & SOLUTIONS

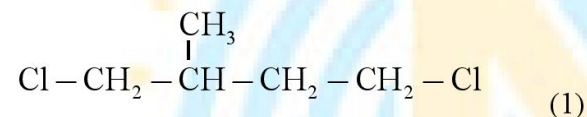
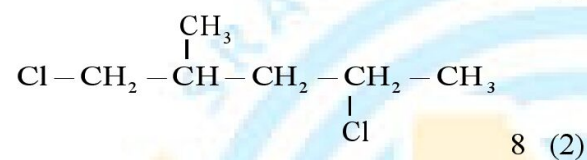
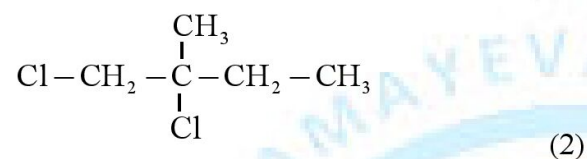
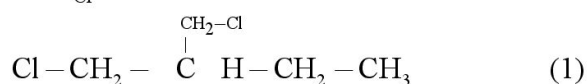
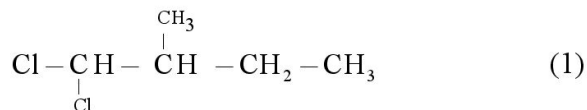
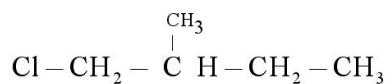
1	D	2	A	3	C	4	D	5	A
6	D	7	C	8	B	9	C	10	B
11	D	12	B	13	C	14	B	15	B
16	B	17	A	18	C	19	D	20	C
21	D	22	C	23	A	24	C	25	D
26	A	27	D	28	B	29	B	30	C
31	C	32	C	33	C	34	B	35	A
36	B	37	D	38	D	39	D	40	B
41	B	42	C	43	B	44	D	45	C
46	A	47	B	48	BC	49	BC	50	ABD
51	ABC	52	AB	53	ABCD	54	B	55	ABD
56	AB	57	BD	58	BD	59	BC	60	ABD
61	ABC	62	CD	63	AD	64	ABCD	65	BC
66	ABCD	67	AC	68	ABCD	69	ACD	70	ABD
71	9	72	4	73	1	74	7	75	4
76	3	77	4	78	3	79	4	80	B
81	C	82	A	83	B	84	D	85	B
86	B	87	C	88	A	89	A	90	B
91	A	92	B	93	C	94	C	95	C
96	D	97	B	98	B	99	A	100	C



SOLUTIONS

5. $3^\circ > 2^\circ > 1^\circ$ follows stability of radicals.

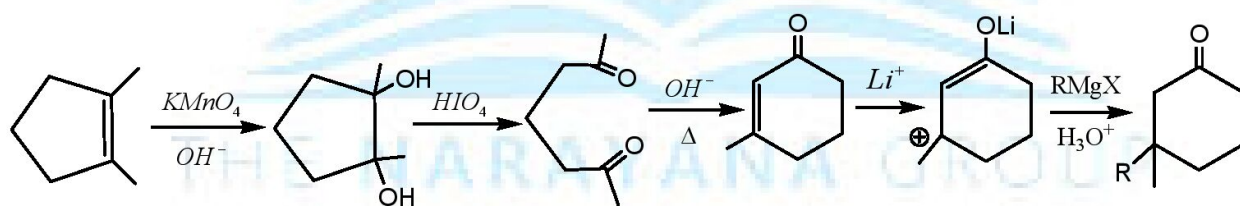
6.



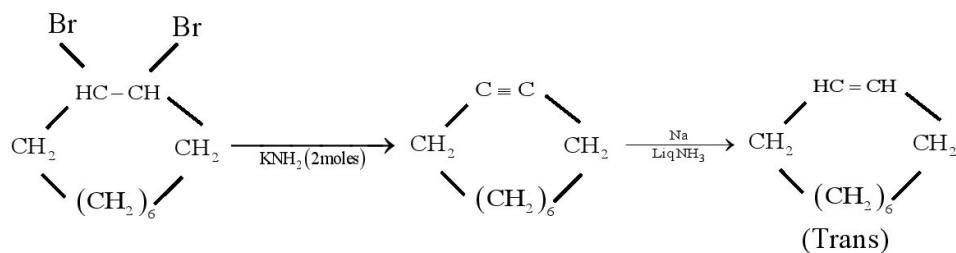
$$7. \frac{\text{Product A}}{\text{Product B}} = \frac{2.6 \times 2}{4 \times 1} = \frac{5.2}{4}$$

$$\therefore \% \text{ A} = \frac{5.2}{9.2} \times 100 = 56.52\%$$

8.



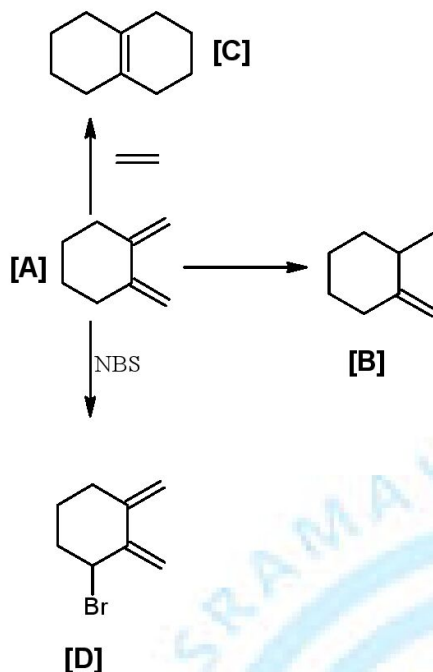
13.



14. $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{C}\equiv\text{CH}$, $\text{CH}_3-\underset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{C}\equiv\text{CH}$

19. Propyne contains acidic -H

20.



21. $CH \equiv CH \xrightarrow{\text{Chromic Acid}} A \xrightarrow{\text{NaOH}} B \xrightarrow[\text{Electrolysis}]{\text{Koble's}} C \xrightarrow[\text{absence of air}]{450^\circ C} D$. Regarding 'D'

$A = CH_3COOH$; $B = CH_3COONa$; $C = CH_3-CH_3$; $D = CH_2=CH_2$

23. Hydroboration followed by oxidation results in the addition a molecule of H_2O

In hydroboration oxidation

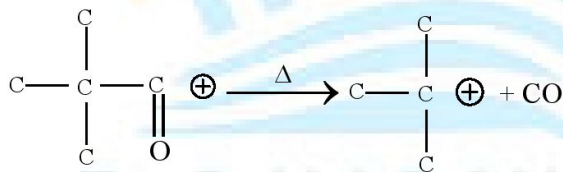
Anti markovnikov product is formed

Syn addition

26. High hyper conjugation

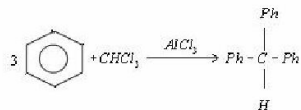
27. Electron releasing groups on aromatic ring activates Electrophilic aromatic substitution reaction

28.



31. Br_2 & Fe

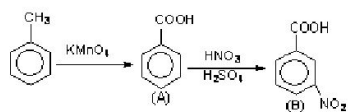
32. 3 moles of C_2H_2 gives 1 mole of C_6H_6 . 0.5 moles gives $\frac{1}{6}$ moles of benzene. 0.5 moles of C_2H_2 obtained by the hydrolysis of 1 mole of 50% pure CaC_2



33.



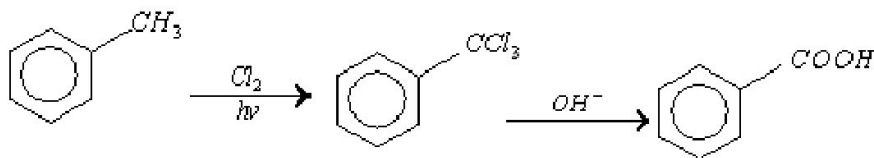
34.



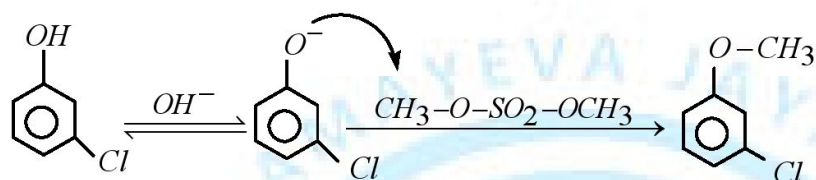
35.

36. During formation of C^{++} most stable is preferred as intermediate C^{++} which attaches to Benzene.
 $3^\circ > 2^\circ > 1^\circ$

37.

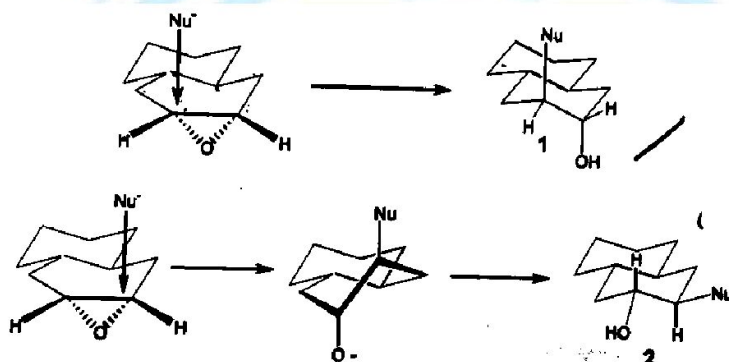


38.



39. $-OCH_3$ group is +M.E group which activates the benzene ring.

43.



In this simple model, the transition-state leading to 1 involves diaxial opening of the epoxide ring. However, formation of diequatorial isomer 2 requires a boat (or skew) like transition-state). Therefore, diaxial opening of the epoxide ring to afford the diaxial product 1 is preferred.

44. Cis +cis addition meso compound

45. Anti addition

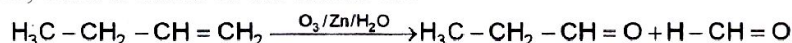
47. Higher the stability of the free radical formed lesser is the dissociation energy. Stability order of the radicals is $H_2C=CH-CH_2\cdot > PhH_2C\cdot > Me_3C\cdot > Me_2C\cdot$.

49. In compound (B) and (C) chiral centres are present after hydrogenation and hydrogenation is syn addition.

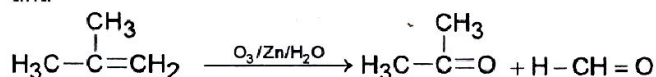
50. 3° Alkyl halides gives elimination reaction with $LiAlH_4$

51. Alkene which on reductive ozonolysis given only ethanal is 2-butene.

So, other isomers of this alkene are

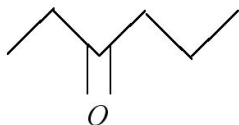


and



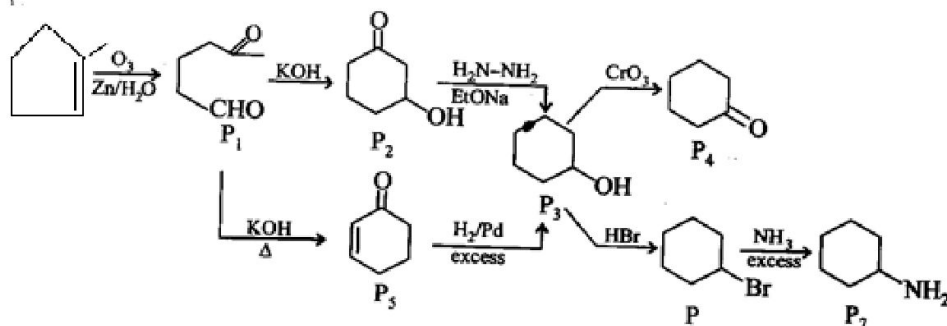
Hence, (A), (B) and (C) is correct answer.

53. Methyl Ketones are formed from terminal alkynes.

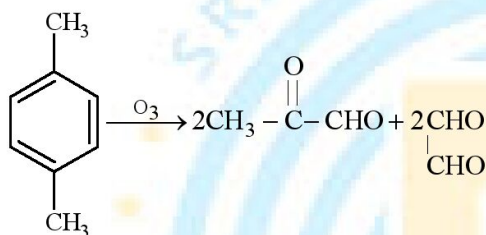


is obtained from either

54.

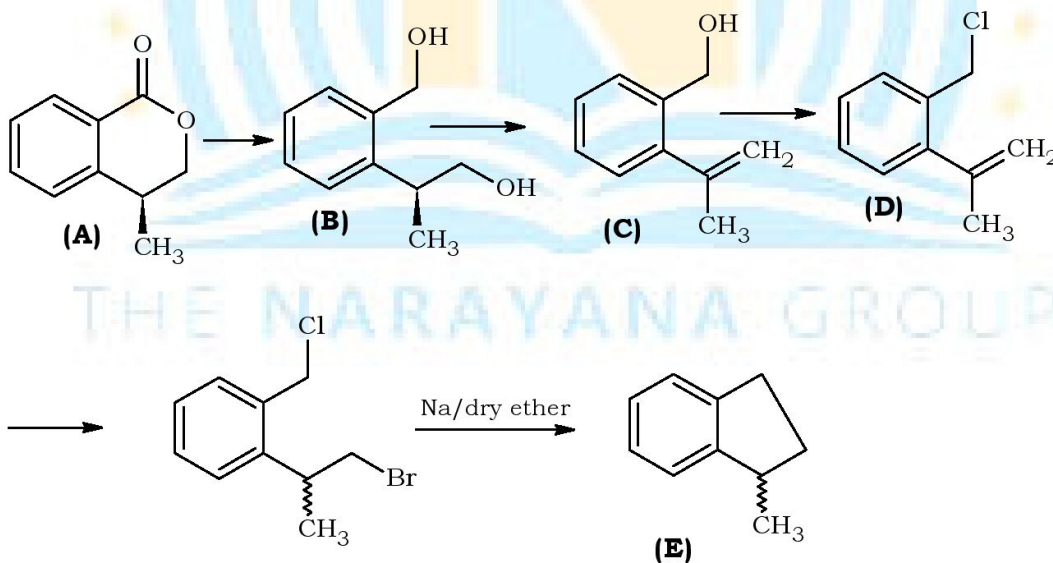


56.



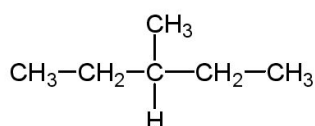
60. (14) Annulenes are aromatic B is more aromatic than A

67.

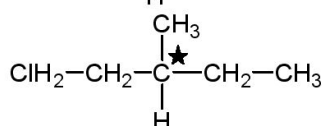


70. Diastereomers are not mirror images of each other. Whereas enantiomers are non-superimposable mirror images.

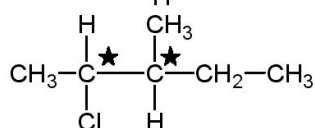
71.



3-methyl pentane monochlorination, different products are

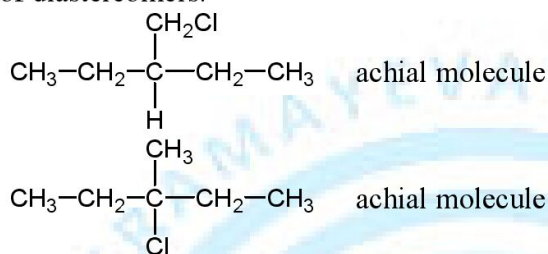


Chiral, one chiral carbon atom pair of enantiomers



Chiral carbon atoms 4 isomers. 2 pairs of enantiomers

4 pairs of diastereomers.



X = number of pairs of enantiomers = 3

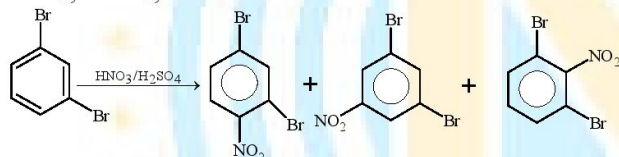
Y = number of pairs of diastereomers = 4

Z = numbers of isomers which are achiral = 2

Total X + Y + Z = 9

73. pent-1-ene

75. $-\text{NR}_2$, $-\text{NH}_2$, $-\text{OH}$ and $-\text{OR}$ are considered as strong activators.



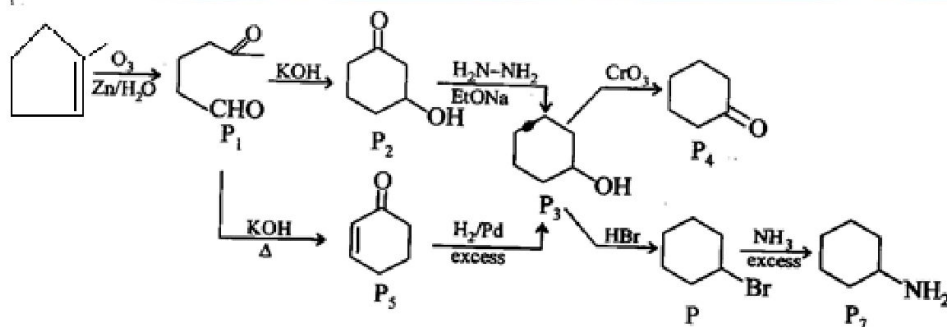
76.

77. a,b,e,g

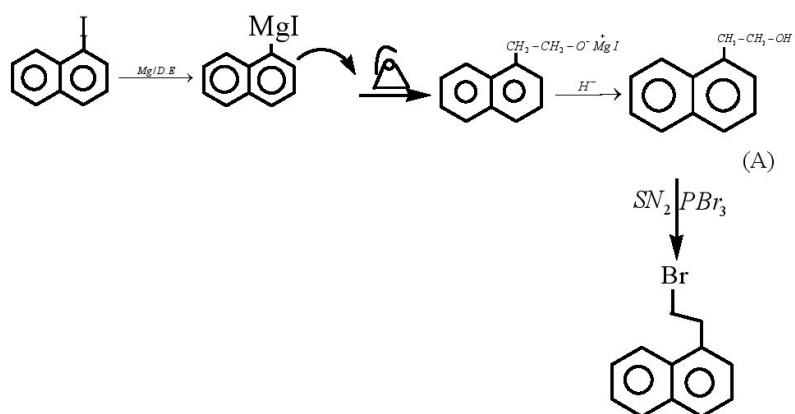
80. $-\text{NO}_2$ group highly deactivate Benzene ring.

81. Benzene ring is highly deactivated due to $-\text{Cl}$ group.

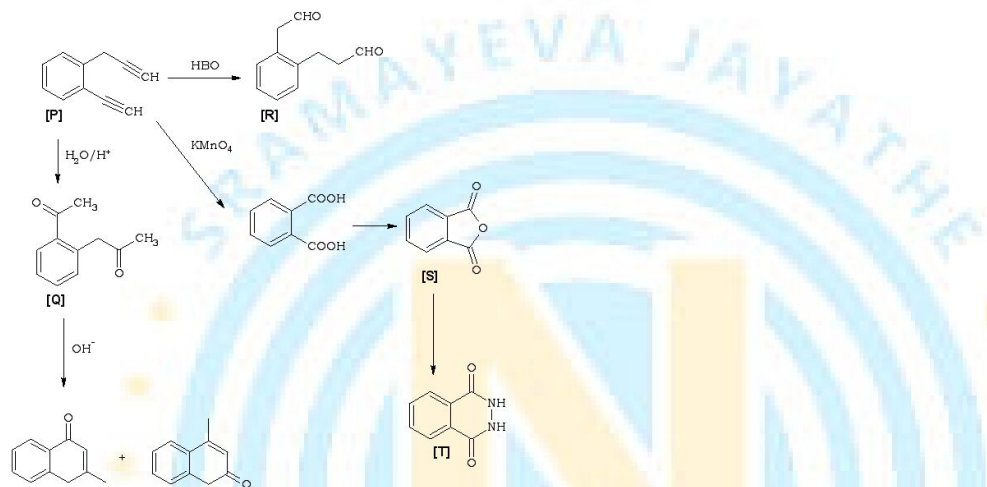
82.



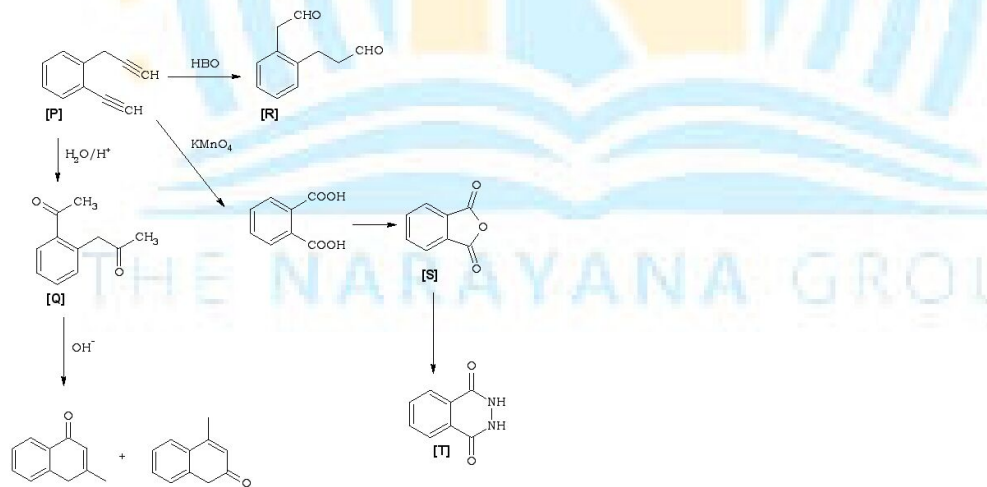
86.



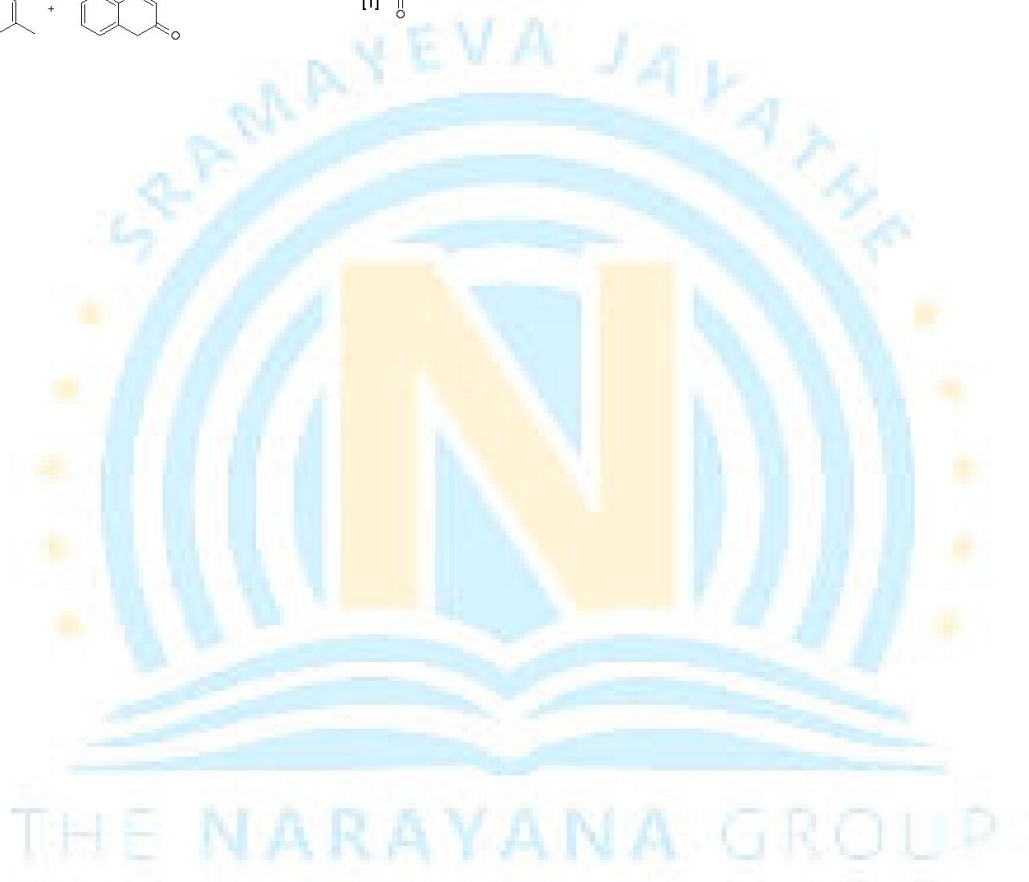
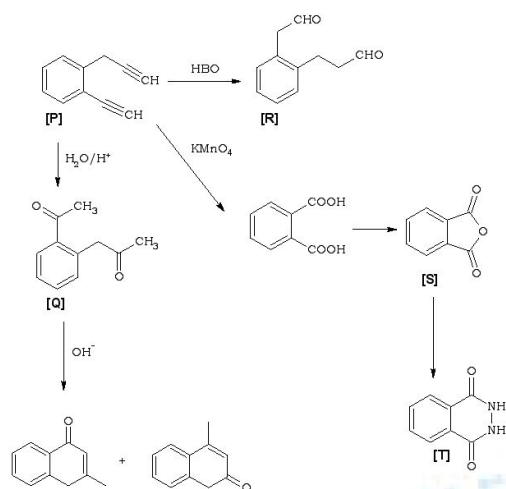
88.



89.



90.





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